EXPERIMENT-10

16-BIT SUBTRACTION

AIM: To write an assembly language program to implent 16 bit subtraction using 8086 processor.

ALGORITHM:

1. Start the program by loading a register pair with address of first number.
2. Copy the data to another register pair.
3. load the second number to first register pair.
4. Subtract the two register pair contacts.
5. Check for borrow.
6. Store the value of difference and borrow in memory location.
7. End.

Program:

MNEMONICS - EXPLANATON

```
MOV AX,[1100H] - load first number
MOV BX,[1102H] - load second number
SUB AX,BX - Subtract BX from AX
MOV [1200H],AX - store difference in memory.
HLT - Halt program execution.
```

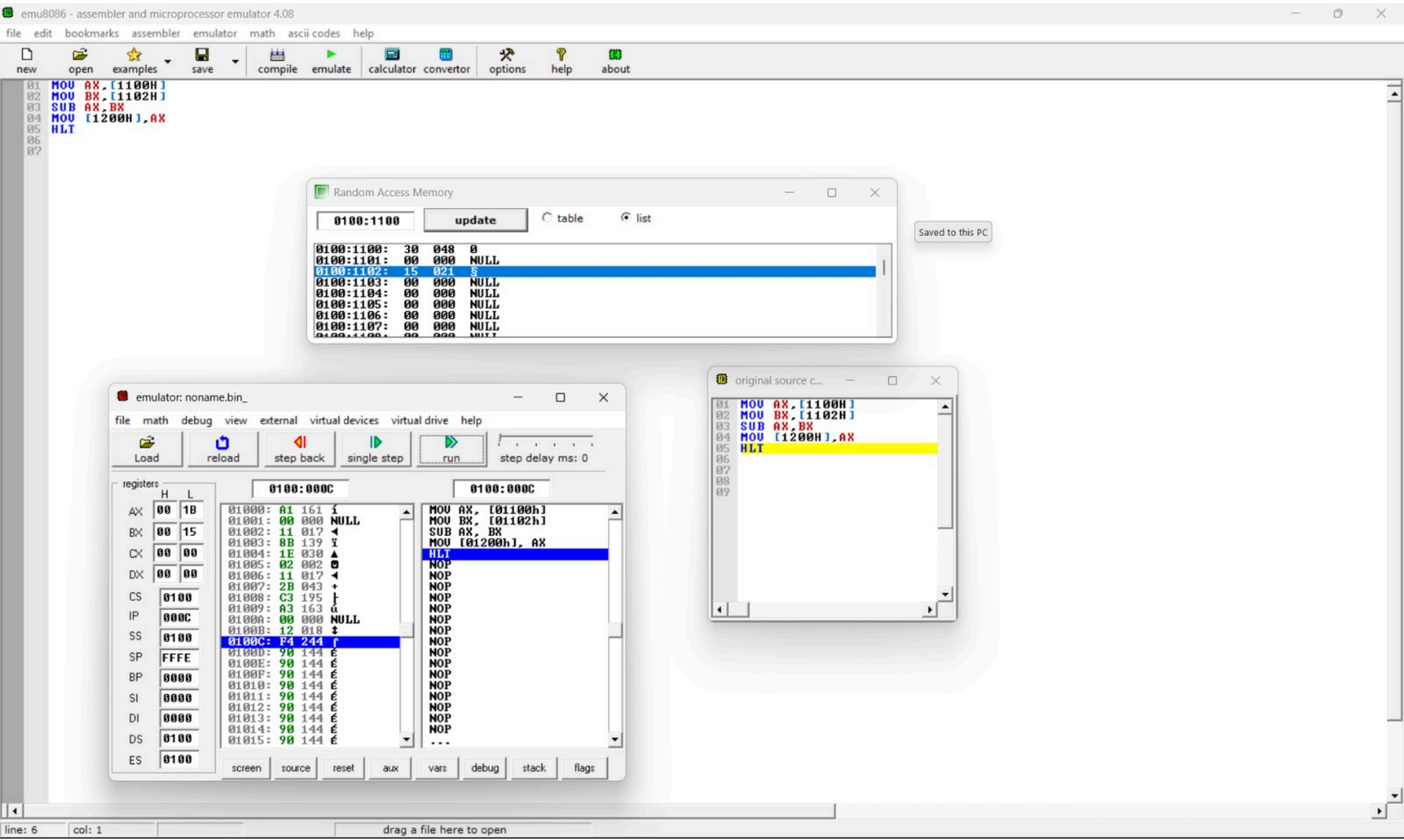
Input:

Address	Data
1100	30
1102	15

Output: O/P ✓

Address	Data
1200	15

Result: Thus the program was executed successfully using 8086 processor simulator.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01 MOV AX,[1100H]
02 MOV BX,[1102H]
03 SUB AX,BX
04 MOV [1200H],AX
05 HLT
Random Access Memory
0100:1100
update
table
list
0100:1100: 30 048 0
0100:1101: 00 000 NULL
0100:1102: 15 021 §
0100:1103: 00 000 NULL
0100:1104: 00 000 NULL
0100:1105: 00 000 NULL
0100:1106: 00 000 NULL
0100:1107: 00 000 NULL
Saved to this PC
original source c...
01 MOV AX,[1100H]
02 MOV BX,[1102H]
03 SUB AX,BX
04 MOV [1200H],AX
05 HLT
emulator: noname.bin_
file math debug view external virtual devices virtual drive help
Load reload step back single step run step delay ms: 0
registers
H L
AX 00 1B
BX 00 15
CX 00 00
DX 00 00
CS 0100
IP 000C
SS 0100
SP FFFE
BP 0000
SI 0000
DI 0000
DS 0100
ES 0100
0100:000C
0100:000C
01000: A1 161 í
01001: 00 000 NULL
01002: 11 017
01003: 8B 139 ï
01004: 1E 030
01005: 02 002
01006: 11 017
01007: 2B 043 +
01008: C3 195
01009: A3 163 ú
0100A: 00 000 NULL
0100B: 12 018
0100C: F4 244
0100D: 90 144 É
0100E: 90 144 É
0100F: 90 144 É
01010: 90 144 É
01011: 90 144 É
01012: 90 144 É
01013: 90 144 É
01014: 90 144 É
01015: 90 144 É
MOV AX, [01100h]
MOV BX, [01102h]
SUB AX, BX
MOV [01200h], AX
HLT
NOP
...
screen source reset aux vars debug stack flags
line: 6 col: 1 drag a file here to open